

Fluid Flow Operation

Text Books:

- **K.A. Gavanhe, “Unit Operations - I”, Nirali Prakashan, 2015**
- W. L. McCabe, J.C. Smith and P. Harriott, “Unit operations of Chemical Engineering”, McGraw Hill, International Edn., 7th Edition, 2005.

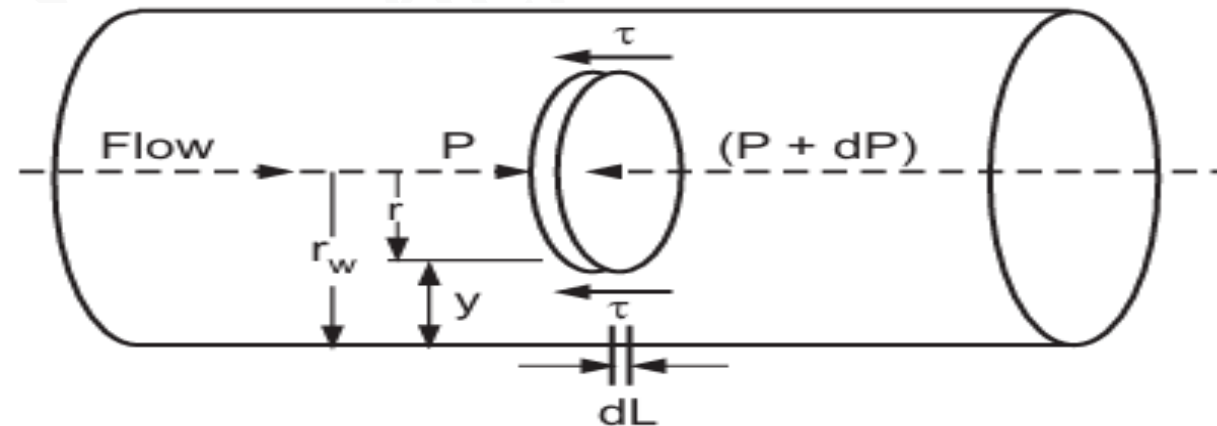
Unit 3: Flow of incompressible fluids in pipes

- Flow of incompressible fluids in pipes
- Laminar and turbulent flow through closed conduits
- Velocity profile & friction factor for smooth and rough pipes
- Head loss due to friction in pipes, fitting etc.

Flow of incompressible fluids in pipes

Shear-stress distribution in a cylindrical tube:

- Steady flow of a fluid through a horizontal tube
- disk-shaped element of fluid, of radius r and length dL , concentric with the axis of the tube
- P and $P + dP$ be the fluid pressures on the upstream and downstream faces of the disk
- Assume fully developed flow and the density of fluid to be constant



Flow of incompressible fluids in pipes

Shear-stress distribution in a cylindrical tube:

- For fully developed flow: sum of all the forces acting on this disk shaped element of fluid is equal to zero.
- Forces acting on disk shaped element:

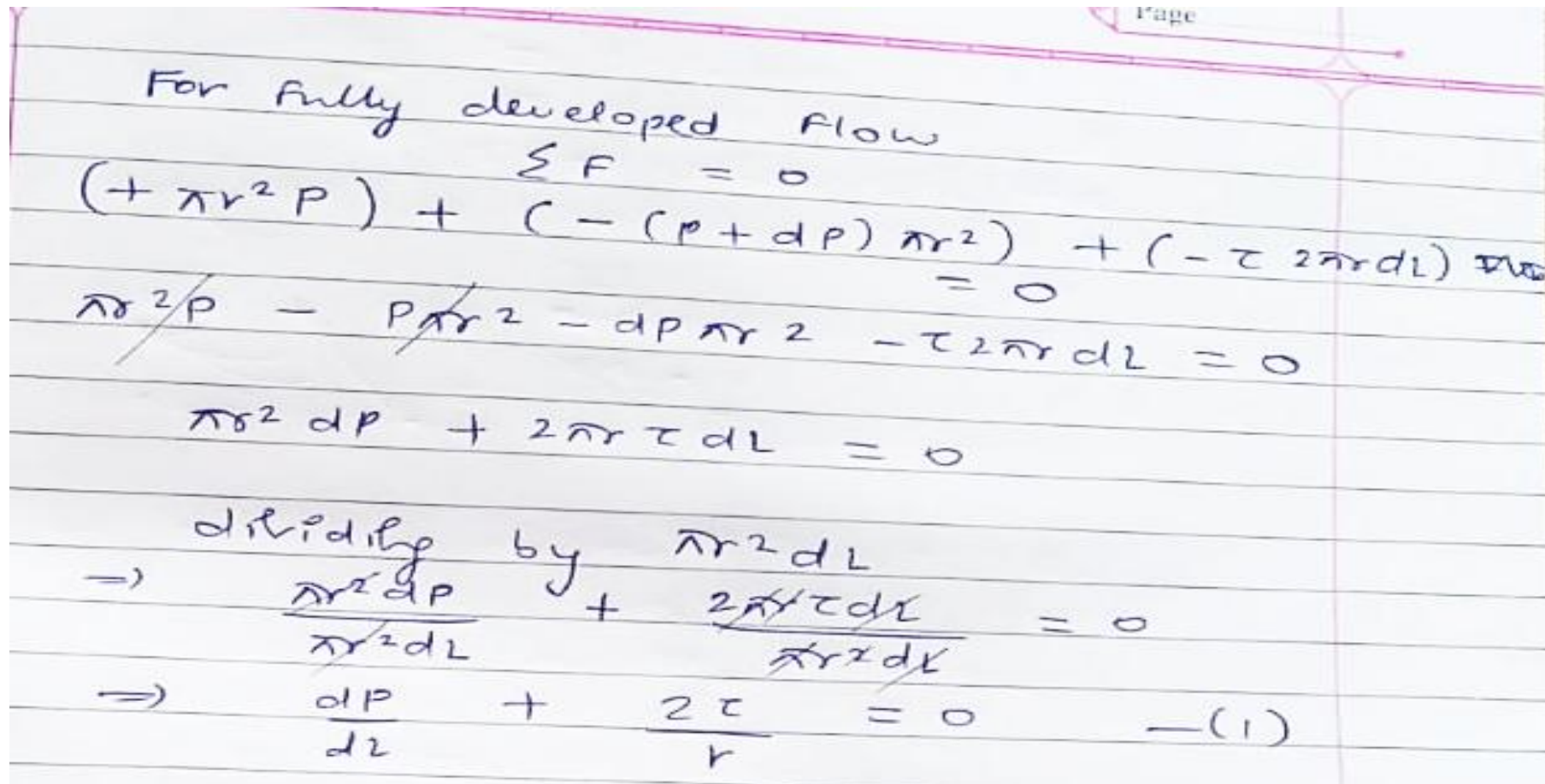
$$\left[\begin{array}{c} \text{Pressure force on the upstream} \\ \text{face of the disk} \end{array} \right] = P\pi r^2 \quad \dots \text{ in the direction of flow (+ve)}$$

$$\left[\begin{array}{c} \text{Pressure force on the downstream} \\ \text{face of the disk} \end{array} \right] = (P + dP) \pi r^2 \quad \dots \text{ opposing the flow (-ve)}$$

$$\left[\begin{array}{c} \text{Shear force opposing flow} \\ \text{acting at the outer surface of the disk} \\ \text{due to the viscosity of fluid} \end{array} \right] = \text{shear stress} \times \text{cylindrical area}$$
$$= \tau \cdot (2\pi r \, dL) \quad \dots \text{ opposing the flow (-ve)}$$

Flow of incompressible fluids in pipes

Shear-stress distribution in a cylindrical tube:



For fully developed flow

$$\sum F = 0$$
$$(+\pi r^2 P) + (- (P + dP) \pi r^2) + (-\tau 2\pi r dL) = 0$$
$$\cancel{\pi r^2 P} - \cancel{P \pi r^2} - dP \pi r^2 - \tau 2\pi r dL = 0$$
$$\pi r^2 dP + 2\pi r \tau dL = 0$$

dividing by $\pi r^2 dL$

$$\Rightarrow \frac{\pi r^2 dP}{\pi r^2 dL} + \frac{2\pi r \tau dL}{\pi r^2 dL} = 0$$
$$\Rightarrow \frac{dP}{dL} + \frac{2\tau}{r} = 0 \quad \text{--- (1)}$$

Flow of incompressible fluids in pipes

Shear-stress distribution in a cylindrical tube:

For steady state laminar / turbulent flow
pressure at any cross section is constant.
 $\frac{dp}{dz}$ independent of r

by taking $\tau = \tau_w$ at $r = r_w$
 τ_w = shear stress at wall of tube
 r_w = radius of tube

$$\therefore \frac{dp}{dz} + \frac{2\tau_w}{r_w} = 0 \quad \text{--- (2)}$$
$$(1) - (2) \Rightarrow$$
$$\frac{dp}{dz} + \frac{2\tau_w}{r_w} - \frac{dp}{dz} - \frac{2\tau}{r} = 0$$
$$\Rightarrow \boxed{\frac{\tau_w}{r_w} = \frac{\tau}{r}}$$

For $r = 0 \Rightarrow \tau = 0 \Rightarrow$ shear stress
is 0 at center of tube

Flow of incompressible fluids in pipes

Shear-stress distribution in a cylindrical tube:

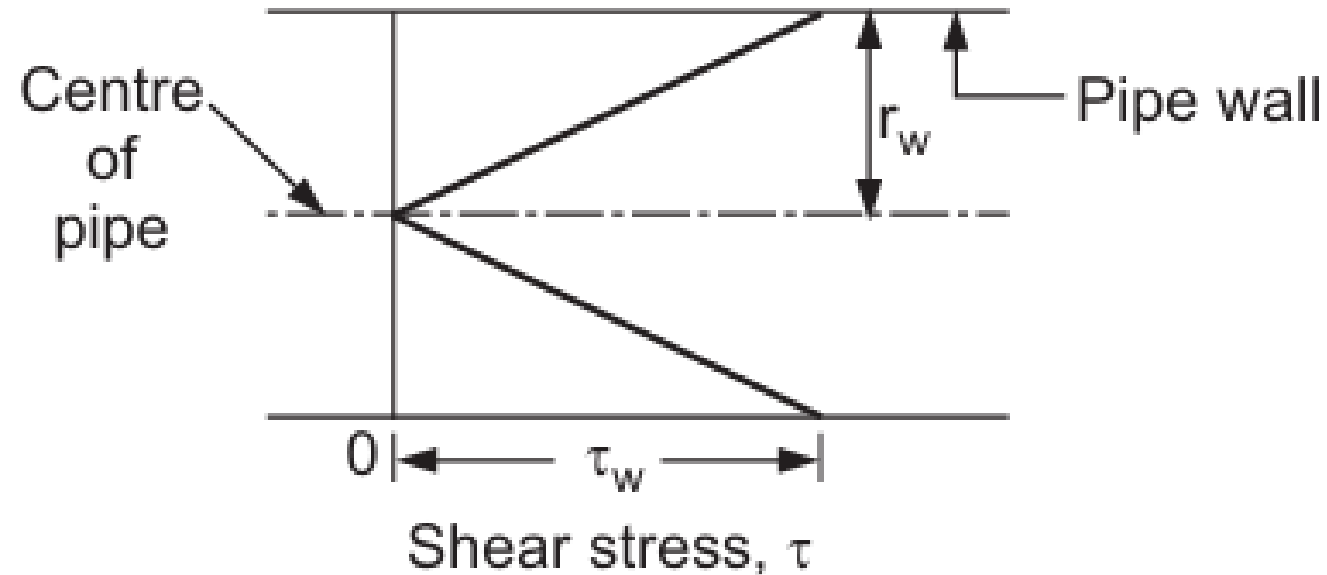


Fig. 7.15 : Shear stress variation in pipe

Flow of incompressible fluids in pipes

- **Skin friction:** The friction that exists between the wall and the fluid stream (h_{fs}).
- Skin friction is the tangential friction associated with a fluid flowing over a smooth surface.
- When a fluid is flowing through a pipe, it is only the skin friction that exists, where D is the diameter of the pipe ($D = 2r_w$).

$$h_{fs} = \frac{\Delta P}{\rho}$$

$$\left[\begin{array}{l} \text{Head loss due to friction} \\ \text{(frictional head loss)} \end{array} \right] = h'_{fs} = \frac{h_{fs}}{g} = \frac{\Delta P}{\rho g}$$

$$\frac{dP}{dL} + \frac{2\tau_w}{r_w} = 0 \quad \longrightarrow \quad h_{fs} = \frac{2\tau_w}{\rho r_w} \Delta L = \frac{4\tau_w}{\rho D} \Delta L$$

The Fanning Friction Factor (f)

- It is defined as the ratio of the wall shear stress to the product of the kinetic energy of fluid and the density.
- It is especially useful in the study of turbulent flow.
- f is a dimensionless and unitless quantity.

$$f = \frac{\tau_w}{\rho \cdot u^2/2} = \frac{2 \tau_w}{\rho u^2}$$

Rearranging

$$\tau_w = f \rho u^2/2$$

The Fanning Friction Factor (f)

$$\begin{aligned} h_{fs} &= \frac{2 \tau_w \Delta L}{\rho r_w} \\ h_{fs} &= \frac{4 \tau_w \Delta L}{\rho D} \\ (1) \quad h_{fs} &= \frac{\Delta P}{\rho} = \frac{4 \tau_w \Delta L}{\rho D} \\ \tau_w &= \frac{f \rho u^2}{2} \\ \Rightarrow h_{fs} &= \frac{4 \left(\frac{f \rho u^2}{2} \right) \Delta L}{\rho D} \\ h_{fs} &= \frac{4 f u^2 \Delta L}{2 D} \end{aligned}$$

The Fanning Friction Factor (f)

ΔL replaced by length of pipe L

$\therefore h_f = \frac{4fu^2L}{2D}$

$h_f = \frac{\Delta P}{\rho g} = \frac{4fu^2L}{2D}$

$\Rightarrow \Delta P = \frac{4f\rho u^2L}{2D}$

Fanning equation: pressure drop due to friction in a pipe for turbulent flow can be calculated from Equation.

Laminar Flow In Circular Pipe

Newton's Law of viscosity

$$\tau = -\mu \frac{du}{dr}$$

$$\mu = \frac{-\tau}{du/dr}$$

Negative sign due to decrease in velocity with increase in radius

Rearranging

$$\frac{du}{dr} = -\frac{\tau}{\mu}$$

$$\frac{\tau_w}{r_w} = \frac{\tau}{r} \Rightarrow \frac{du}{dr} = -\frac{\tau_w}{r_w \mu} \cdot r$$

$$\frac{du}{dr} = -\frac{\tau_w}{r_w \mu} \cdot r$$

u

r

Laminar Flow In Circular Pipe

$$\int_0^u du = -\frac{\tau_w}{r_w \mu} \int_{r_w}^r r dr$$
$$u = \frac{\tau_w}{r_w \mu} \left[\frac{r^2}{2} \right]_{r_w}^r$$
$$\Rightarrow u = \frac{\tau_w}{2 r_w \mu} [r_w^2 - r^2]$$

At center of pipe $r=0$ $u = u_{\max}$
(maximum velocity at center of pipe)

$$u_{\max} = \frac{\tau_w}{2 r_w \mu} r_w^2$$
$$\therefore u_{\max} = \frac{\tau_w \cdot r_w}{2 \mu}$$
$$\tau_w = \frac{\Delta P \cdot r_w}{2 \Delta L} \quad \rightarrow \quad u_{\max} = \frac{\Delta P r_w}{2 \Delta L} \times \frac{r_w}{2 \mu}$$
$$u_{\max} = \frac{\Delta P r_w^2}{4 \Delta L \mu}$$

Laminar Flow In Circular Pipe

$$D = 2r_w \Rightarrow u_{max} = \frac{\Delta P \cdot D^2}{16\mu \cdot \Delta L}$$

Divide u by u_{max}

$$\frac{u}{u_{max}} = \frac{\frac{\tau_w}{2r_w} (r_w^2 - r^2)}{\frac{\tau_w r_w}{2\mu}}$$
$$= \frac{r_w^2 - r^2}{r_w^2}$$
$$\frac{u}{u_{max}} = \left[1 - \left(\frac{r}{r_w} \right)^2 \right]$$

Laminar Flow In Circular Pipe

$$u = u_{max} \left[1 - \left(\frac{r}{r_w} \right)^2 \right]$$

Equation for velocity profile across pipe.

velocity profile will be a parabola with apex at centreline of pipe.

Average velocity u of entire stream through cross section A

$$u = \frac{1}{A} \int_A u \cdot dA$$

Laminar Flow In Circular Pipe

$$\begin{aligned} A &= \pi r^2 \\ dA &= 2\pi r dr \\ \text{area of ring of radius } r \text{ and width } dr \\ A &= \pi r_w^2 \quad (\text{at pipe wall}) \\ u &= \frac{\tau_w}{2r_w \mu} [r_w^2 - r^2] \\ u &= \frac{\tau_w}{2r_w \mu} \times \frac{2\pi}{\pi r_w^2} \int_0^{r_w} (r_w^2 - r^2) r dr \\ u &= \frac{\tau_w}{r_w^3 \mu} \int_0^{r_w} (r_w^2 - r^2) r dr \\ &= \frac{\tau_w}{r_w^3 \mu} \left[\int_0^{r_w} r_w^2 r dr - \int_0^{r_w} r^3 dr \right] \\ &= \frac{\tau_w}{r_w^3 \mu} \left[r_w^2 \left[\frac{r^2}{2} \right]_0^{r_w} - \left[\frac{r^4}{4} \right]_0^{r_w} \right] \end{aligned}$$

Laminar Flow In Circular Pipe

$$\begin{aligned} &= \frac{\tau_w}{r_w^3 \mu} \left[r_w^2 \times \frac{r_w^2}{2} - \frac{r_w^4}{4} \right] \\ &= \frac{\tau_w}{r_w^3 \mu} \left[\frac{2r_w^4 - r_w^4}{4} \right] \\ &= \frac{\tau_w \times r_w^4}{4 r_w^3 \mu} \\ \Rightarrow u &= \frac{\tau_w r_w}{4 \mu} \\ \frac{u}{u_{\max}} &= \frac{\tau_w r_w / 4 \mu}{\tau_w r_w / 2 \mu} = \frac{1}{2} = 0.5 \\ \Rightarrow \text{Average velocity is half of maximum velocity.} \end{aligned}$$

Laminar Flow In Circular Pipe

$$u = \frac{\tau_w r_w}{4\mu}$$
$$\left[\begin{aligned} \tau_w &= \frac{\Delta P r_w}{2\Delta L} & r_w &= \frac{D}{2} \\ \rightarrow u &= \frac{\Delta P r_w}{2\Delta L} \times \frac{r_w}{4\mu} \\ &= \frac{\Delta P D}{2 \times 2\Delta L} \times \frac{D}{2 \times 4\mu} \\ u &= \frac{\Delta P D^2}{32\mu\Delta L} \end{aligned} \right.$$
$$\Rightarrow \Delta P = \frac{32\Delta L \mu u}{D^2}$$

replacing ΔL by L - length of pipe

$$\Rightarrow \boxed{\Delta P = \frac{32 L \mu u}{D^2}}$$

Equation for pressure drop over L length of pipe.

Laminar Flow In Circular Pipe

- Hagen-Poiseuille equation: provides ΔP is the pressure drop over the length L of the pipe.

$$\Delta P = \frac{32 L \mu u}{D^2}$$

- The Hagen-Poiseuille equation is useful to determine experimentally the viscosity of a fluid by measuring the pressure drop and the volumetric flow rate through a tube of a given length and diameter.
- This equation also useful for the calculation of the pressure drop due to friction in laminar/viscous flow if the viscosity is known.

Laminar Flow In Circular Pipe

$$\text{Fanning Friction Factor } f = \frac{2\tau_w}{\rho u^2}$$

$$u = \frac{\tau_w r_w}{4\mu} = \frac{\tau_w D}{8\mu}$$

$$\tau_w = \frac{8\mu u}{D}$$

$$\Rightarrow f = \frac{2}{\rho u^2} \times \frac{8\mu u}{D}$$

$$= \frac{16\mu}{\rho u D}$$

$$= \frac{16}{\left(\frac{\rho u D}{\mu}\right)}$$

$$f = \frac{16}{N_{Re}}$$

for laminar flow
through pipes

Turbulent Flow In Pipes

- The turbulent flow is characterised by the presence of eddies (eddy - a circular movement of, say, water causing a small whirlpool).
- The eddies are of various sizes and they coexist.
- The fluctuations/oscillations are present in all the three directions.
- The fluctuations cause the mixing of different fluid portions by lateral movement.
- The turbulent flow is of a great importance because it brings about mixing of the fluid elements which results in higher rates of heat and mass transfer.

Turbulent Flow In Pipes

- Fanning friction factor as a function of the Reynolds number for turbulent flow through a smooth pipe is given by Nikuradse equation.

$$\frac{1}{\sqrt{f}} = 4 \log (N_{Re} \sqrt{f}) - 0.40$$

- This equation requires a trial and error procedure for estimating the friction factor.
- The other empirical relation that can be used for estimating the friction factor from the Reynolds number for turbulent flow is

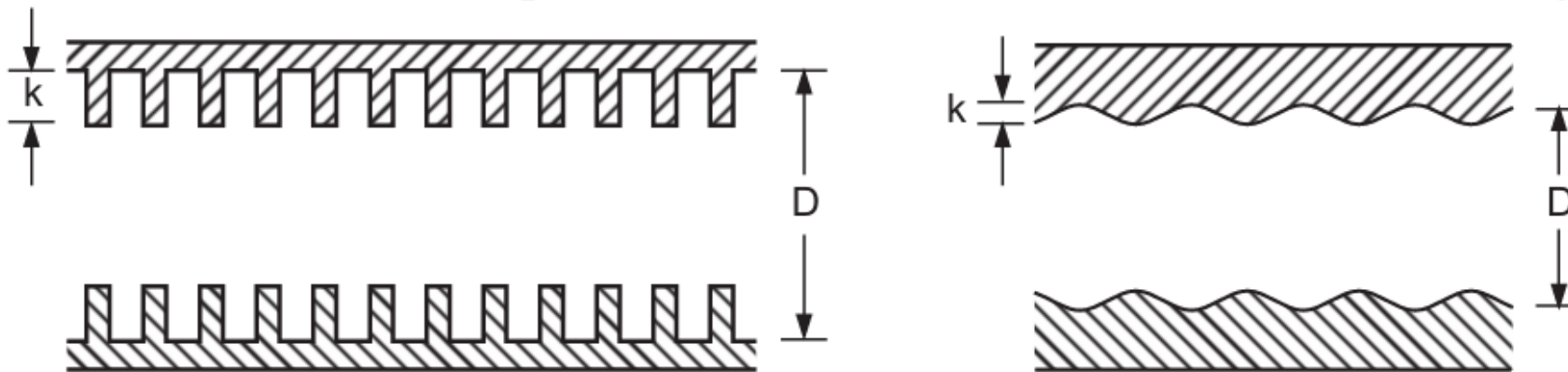
$$f = \frac{0.078}{(N_{Re})^{0.25}}$$

Effect of Roughness

- In turbulent flow, a rough pipe results in a larger friction factor than that with a smooth pipe, for a given Reynolds number.
- If a rough pipe is made smooth, the friction factor is reduced and ultimately a stage will come when further smoothening of the pipe does not reduce the friction factor for a given Reynolds number.
- The pipe is then said to be hydraulically smooth.
- A roughness parameter (k) is a length representing the magnitude of surface roughness.
- Relative roughness is the ratio of the roughness parameter to the diameter of pipe, k/D .
- The friction factor dependency on the surface roughness is given through the relative roughness.

Effect of Roughness

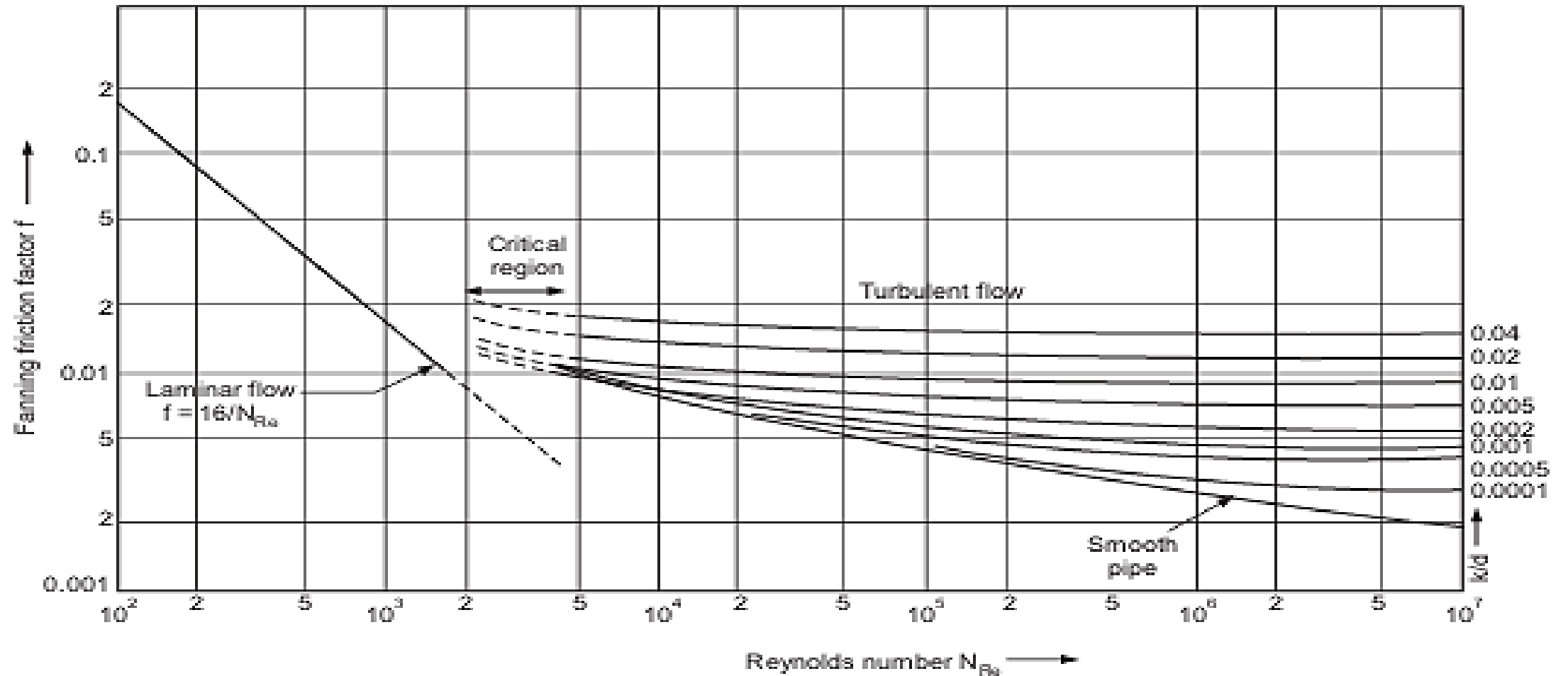
- For low values of Reynolds number ($N_{Re} < 2100$), the friction factor is independent of the surface roughness.
- For large values of Reynolds number ($N_{Re} > 2500$), the friction factor is also a function of the surface roughness.
- At very large values of Reynolds number, the friction factor is independent of N_{Re} but is a function of the surface roughness.



Friction Factor Chart

- It is a logarithmic plot of friction factor f v/s Reynolds number over a wide range of the Reynolds number for flow of fluids in smooth as well as rough pipes.
- The data for the plot are taken over a wide range of each variable such as velocity, density and pipe diameter, etc. using liquids and gases.
- This chart is a graphical representation of the relationship : $f = f(N_{Re}, k/D)$, the relationship between the friction factor and the Reynolds number with k/D as a parameter on log-log paper.
- The friction factor is a function of both the Reynolds number and relative roughness.
- For laminar flow region, the friction factor is not affected by the relative roughness of a pipe, therefore, only one line is shown for Reynolds number upto about 2100.

Friction Factor Chart



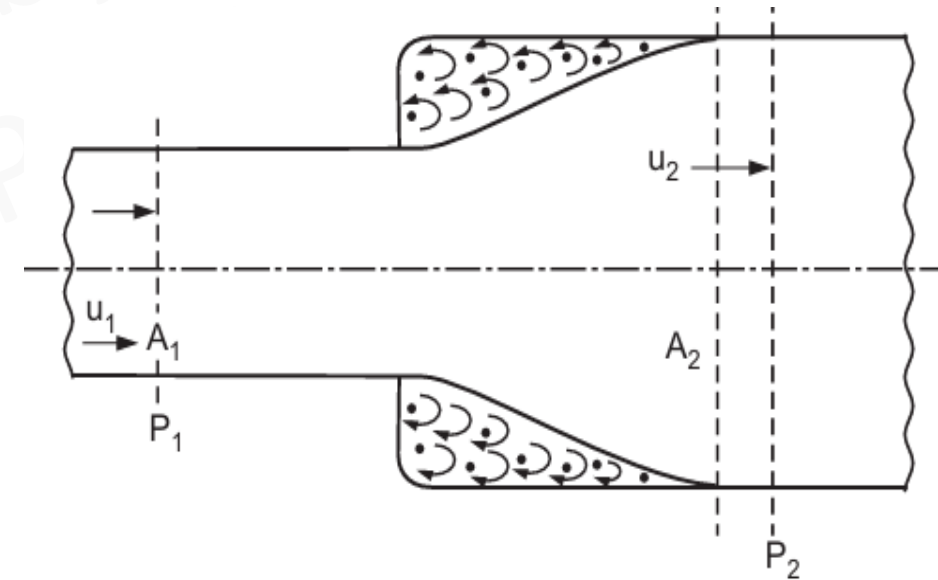
Roughness of pipewall in μm : Steel tube : 40, Cast iron : 300, Galvanised iron : 150

Friction Factor Chart

- For the laminar flow region, the relation $f = 16/N_{Re}$ expresses the dependency of f on N_{Re} .
- A log-log plot of the above relation results in a straight line with a slope equal to minus one.
- For turbulent flow, the relationship between ' f ' and N_{Re} is given by the Nukuradse equation.
- Since ' f ' is also a function of the relative roughness for turbulent flow, curves with different values of the dimensionless ratio k/D are presented in the chart for $N_{Re} > 2100$ assuming that turbulent flow exist at all Reynolds number greater than 2100.
- The friction factor is low for a smooth pipe as compared to a rough pipe and increases as the relative roughness increases for a given Reynolds number.
- For Reynolds number above 4000, the flow is turbulent and values of f should be read from lines at the right, where the lowest line represents the friction factor for smooth pipes.

Friction Loss from Sudden Enlargement:

- If the cross-section of the pipe is suddenly enlarged (increased), there is a friction loss (mechanical energy loss) due to eddies which are greater at the point of sudden enlargement than in the straight pipe.
- The fluid stream separates from the wall of the pipe and issues in the enlarged section as a jet, which then expands and consequently fills the entire cross-section of the larger pipe.
- The space between the expanding jet and the pipe wall is filled with eddies (fluid vortices) and a large amount of friction is generated.



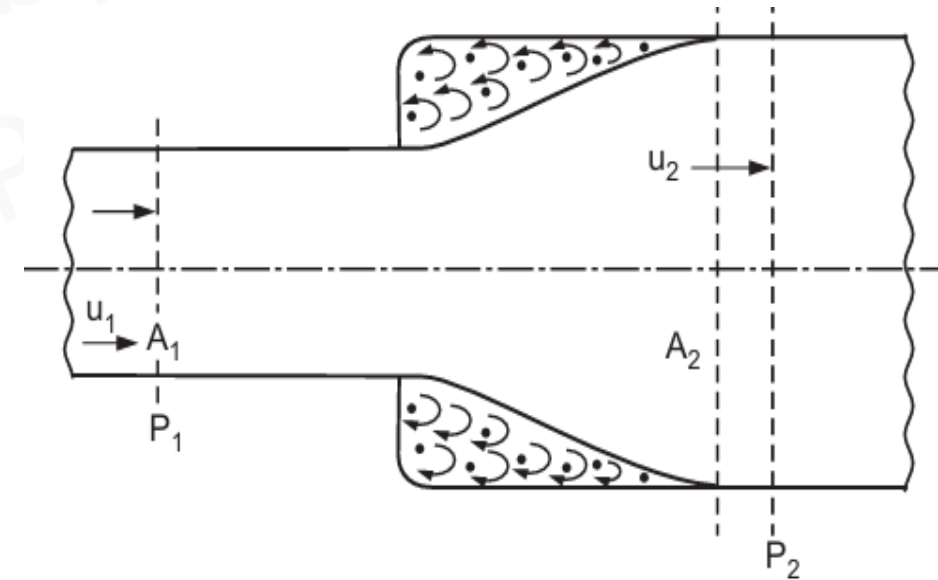
Friction Loss from Sudden Enlargement:

- The friction head loss from a sudden enlargement (expansion) of the cross-section of the pipe for turbulent flow is proportional to the velocity head of the fluid in the smaller pipe and is given by

$$h_{fe} = K_e \cdot \frac{u_1^2}{2g}$$

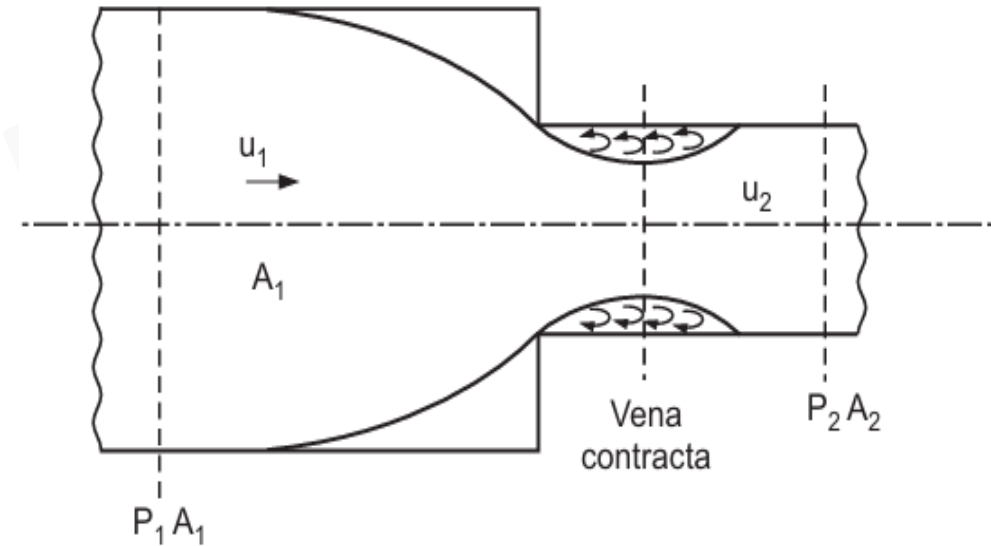
- where K_e is a proportionality constant and known as the expansion-loss coefficient and u_1 is the average velocity of fluid in the smaller pipe/upstream pipe.

$$K_e = \left(1 - \frac{A_1}{A_2}\right)^2$$



Friction Loss from Sudden Contraction

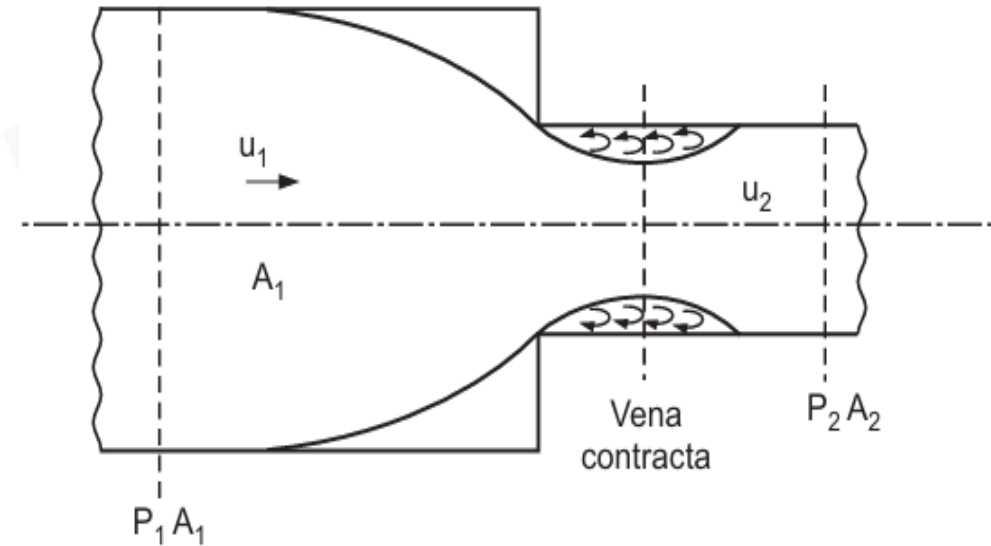
- When the cross-section of the pipe is suddenly reduced, the fluid stream gets separated from the wall of the pipe and a fluid jet is formed.
- This jet first contracts, at a short distance from the sudden contraction, in the smaller pipe and then expands to fill the entire cross-section of the smaller pipe.
- The cross-section of minimum effective flow area at which the fluid jet changes from a contraction to an expansion is known as the vena contracta.
- Upto the vena contracta, the fluid is accelerated and losses are very small.



Friction Loss from Sudden Contraction

- But beyond it, the velocity decreases as the flow area increases and conditions are equivalent to those for a sudden expansion.
- The space between the wall and the jet is filled with eddies.
- The friction loss or frictional head loss from a sudden contraction of the cross-section of the pipe is proportional to the velocity head of the fluid in the smaller diameter pipe.

$$h_{fc} = K_c \frac{u_2^2}{2g}$$



Friction Losses in Fittings and Valves

- Various types of fittings and valves are used in industrial piping systems to change the direction of flow, for connecting pipes of different diameters, etc. and valves are used to control the flow or to stop the flow of a fluid.
- Fittings and valves disturb the normal flow-line and cause friction and may lead to greater frictional loss than that caused by the straight pipe.
- The frictional loss due to fittings and valves may be given either in terms of the velocity head or in terms of the equivalent length.
- The frictional head loss in terms of the velocity head is given by the following equation, where K_f is the loss factor for the fitting and u_1 is the average velocity of the fluid in the piping leading to the fitting.

$$h_{ff} = K_f \frac{u_1^2}{2g}$$

Friction Losses in Fittings and Valves

- The equivalent length of a fitting is that length of straight pipe of the same nominal size as that of the fitting, which would cause the same friction loss as that caused by the fitting.
- The equivalent length of a fitting/valve is usually expressed as a certain number of pipe diameters.
- Thus, if a bend has an equivalent length of X diameters and if the nominal pipe diameter is 5 cm, then the bend will cause the frictional loss equivalent to that caused by the straight pipe of length $5 X$ cm.
- The equivalent length of a fitting is to be added to the length of a straight pipe to get the total equivalent length of a flow system composed of the straight pipe and the fitting.

Friction Losses in Fittings and Valves

Equivalent lengths of some pipe fittings are :

Fitting	Equivalent length, pipe diameters
45° Elbow	15
90° Elbow	32 (upto $D = 40$ mm)
Tee	60 to 90
Coupler and union joint	Negligible
Gate valve (fully open)	10
Globe valve (fully open)	300

Form-friction losses in the Bernoulli equation

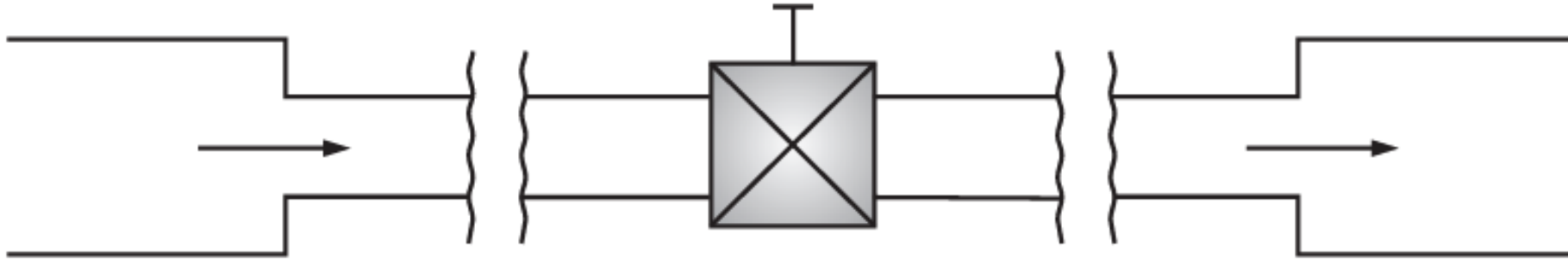


Fig. 7.20 : Typical flow system for incompressible fluid

Total head loss due to friction in m of flowing fluid for the system of Fig. 7.20 is given by

$$h_f = \left[\frac{4 fL}{D} + K_c + K_e + K_f \right] \frac{u^2}{2g} \quad \dots (7.129)$$

where u is the average velocity of the fluid through the pipe of diameter D and f is the Fanning friction factor.

Friction Loss at the Exit from a Pipe

- The outlet end of a pipe carrying fluid may be either left free so that the fluid is discharged freely in the atmosphere or it may be connected to a large reservoir where the pipe outlet is submerged and fluid is discharged into a large body of static fluid.
- In such a case, the loss of head at the exit of the pipe is equal to $u^2/2g$, where u is the average velocity of flow in the pipe.